

DEPARTMENT OF MATHEMATICS, IIT - GUWAHATI

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MA 101 Mathematics I

Problem Sheet 5: Line Integrals and applications, Green's Theorem, Stokes Theorem and Divergence Theorem.

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1. Calculate the line integral of the vector field along the path described:

- (a) $f(x, y) = (x^2 + y^2)\mathbf{i} + (x^2 - y^2)\mathbf{j}$ from $(0, 0)$ to $(2, 0)$ along the curve $y = 1 - |1 - x|$
 (b) $f(x, y, z) = 2xy\mathbf{i} + (x^2 + z)\mathbf{j} + (y + z)\mathbf{k}$ from $(1, 0, 2)$ to $(3, 4, 1)$ along a line segment
 (c) $f(x, y, z) = x\mathbf{i} + y\mathbf{j} + (xz - y)\mathbf{k}$, along the path described by $\alpha(t) = t^2\mathbf{i} + 2t\mathbf{j} + 4t^3\mathbf{k}$, $0 \leq t \leq 1$.

Solution:

(a) Note that the parametric representation of the curve $\mathbf{r}(x)$ is given by:

$$\mathbf{r}(x) = x\mathbf{i} + x\mathbf{j} \text{ for } 0 \leq x \leq 1,$$

$$\mathbf{r}(x) = x\mathbf{i} + (2 - x)\mathbf{j} \text{ for } 1 \leq x \leq 2.$$

$$\begin{aligned} \int_C f \cdot d\mathbf{r} &= \int_C f(\mathbf{r}(x)) \cdot \mathbf{r}'(x) dx = \int_0^1 ((x^2 + x^2)1 + (x^2 - x^2)1) dx \\ &+ \int_1^2 ((x^2 + (2 - x)^2)1 + (x^2 - (2 - x)^2)(-1)) dx \\ &= \frac{4}{3}. \end{aligned}$$

(b) Note that the parametrization of the line is given by:

$$\mathbf{r}(t) = \langle 1, 0, 2 \rangle + t(\langle 3, 4, 1 \rangle - \langle 1, 0, 2 \rangle) = \langle 2t + 1, 4t, 2 - t \rangle \text{ for } 0 \leq t \leq 1.$$

Hence the line integral is given by:

$$\begin{aligned} \int_C f \cdot d\mathbf{r} &= \int_C f(\mathbf{r}(t)) \cdot \mathbf{r}'(t) dt = \int_0^1 (2(2t + 1)(4t)2 + ((2t + 1)^2 + 2 - t)4 + (4t + 2 - t)(-1)) dt \\ &= 40 - \frac{3}{2} = \frac{77}{2}. \end{aligned}$$

(c) The line integral is given by:

$$\int_C f \cdot d\mathbf{r} = \int_0^1 f(\mathbf{r}(t)) \cdot \mathbf{r}'(t) dt = \int_0^1 ((t^2)2t + (2t)2 + (4t^5 - 2t)12t^2) dt = \frac{5}{2}.$$

2. Find the line integral of $f(x, y, z) = z$ with respect to arc length of the curve given by $\mathbf{r}(t) = (t \cos t)\mathbf{i} + (t \sin t)\mathbf{j} + t\mathbf{k}$, $0 \leq t \leq 1$.

$$\begin{aligned} \text{Solution: } \int_C f ds &= \int_0^1 t |\mathbf{r}'(t)| dt = \int_0^1 t \sqrt{(\cos t - t \sin t)^2 + (\sin t + t \cos t)^2 + 1} dt \\ &= \int_0^1 t \sqrt{2 + t^2} dt = \frac{1}{3} (3^{\frac{3}{2}} - 2^{\frac{3}{2}}). \end{aligned}$$

3. For each of the following vector fields show that $\mathbf{f}$ is not a gradient vector in $\mathbf{R}^2$. Then for each of the following find a closed path C such that $\oint_C \mathbf{f} \neq 0$ and if possible find a closed path C such that $\oint_C \mathbf{f} = 0$. ($\oint_C \mathbf{f}$ is also used to represent $\oint_C \mathbf{f} \cdot d\mathbf{r}$.)

(a) $\mathbf{f}(x, y) = y\mathbf{i} - x\mathbf{j}$

(b) $\mathbf{f}(x, y) = \frac{y}{(x^2 + y^2)}\mathbf{i} - \frac{x}{(x^2 + y^2)}\mathbf{j}$, for $(x, y) \neq (0, 0)$.

Solution: (a) Since $\frac{\partial P}{\partial y}$ and $\frac{\partial Q}{\partial x}$ (where $P(x, y) = y$ and $Q(x, y) = -x$) are continuous functions

but $\frac{\partial P}{\partial y} = 1 \neq \frac{\partial Q}{\partial x} = -1$, $\mathbf{f}$ is not a gradient vector.

A necessary condition for $\oint_C \mathbf{f}$ to be equal to 0 for every closed path C in $\mathbf{R}^2$ is:

$$\frac{\partial P}{\partial y} = \frac{\partial Q}{\partial x} \text{ for all } (x, y) \in \mathbf{R}^2 \text{ (or } \mathbf{f} \text{ is the gradient vector of some scalar function).}$$

In this case since $\frac{\partial P}{\partial y} = 1 \neq \frac{\partial Q}{\partial x} = -1$ for all $(x, y) \in \mathbf{R}^2$, so for all smooth simple closed curves C in $\mathbf{R}^2$, $\oint_C \mathbf{f} \neq 0$.

Take for example a positively oriented circle C of radius $r > 0$ centered at the origin,

$$C = r \cos t \mathbf{i} + r \sin t \mathbf{j}, \quad 0 \leq t \leq 2\pi$$

$$\text{then } \oint_C \mathbf{f} = \int_0^{2\pi} (r \sin t(-r \sin t) - r \cos t(r \cos t)) dt = -2\pi r^2 \neq 0.$$

(You can also use Green's theorem to check this).

$$(b) \quad \frac{\partial P}{\partial y}, \frac{\partial Q}{\partial x} \quad \left(= \frac{x^2 - y^2}{(x^2 + y^2)^2} \right) \quad \text{are continuous for all } (x, y) \in \mathbf{R}^2 \setminus \{(0, 0)\},$$

$$\text{where } P(x, y) = \frac{y}{(x^2 + y^2)} \quad \text{and } Q(x, y) = -\frac{x}{(x^2 + y^2)}.$$

Since $\mathbf{f}$ is not defined at $(0, 0)$, so $\mathbf{f}$ is not the gradient vector of any scalar function defined throughout $\mathbf{R}^2$, but since $\frac{\partial P}{\partial y} = \frac{\partial Q}{\partial x}$ for all $(x, y) \neq (0, 0)$,

$\mathbf{f}$ is the gradient vector of some scalar function in any open simply connected subset of $\mathbf{R}^2$ not containing the origin.

If we take C as the positively oriented unit circle centered at $(2, 0)$ such that the closed disc enclosed by C does not contain the origin then check that $\oint_C \mathbf{f} = 0$.

(You can also use Green's theorem to check this).

Whereas if we choose C as the positively oriented unit circle centered at the origin, that is $C = \cos t \mathbf{i} + \sin t \mathbf{j}$, $0 \leq t \leq 2\pi$ then

$$(**) \quad \oint_C \mathbf{f} = \int_0^{2\pi} (\sin t(-\sin t) - \cos t(\cos t)) dt = -2\pi \neq 0 \Rightarrow \mathbf{f} \text{ is not a gradient vector in } \mathbf{R}^2 - \{(0, 0)\}.$$

4. Show that each of the following functions F is a gradient vector and find an f for each F such that $F = \nabla f$.

$$(a) \quad F(x, y) = 3x^2y \mathbf{i} + x^3 \mathbf{j}$$

$$(b) \quad F(x, y) = (\sin y - y \sin x + x) \mathbf{i} + (\cos x + x \cos y + y) \mathbf{j}.$$

Solution:

(a) Since P, Q are continuously differentiable functions, (where $P(x, y) = 3x^2y$ and $Q = x^3$)

and $\frac{\partial P}{\partial y} = 3x^2 = \frac{\partial Q}{\partial x}$, F is the gradient of some scalar function f .

Note that if f is such that F is the gradient vector of f , then

$$\frac{\partial f}{\partial x} = P(x, y) = 3x^2y \quad \text{and} \quad \frac{\partial f}{\partial y} = Q(x, y) = x^3, \text{ hence}$$

$$f(x, y) = \int P(x, y) dx + g(y), \text{ where } g \text{ is independent of } x.$$

$$\text{Also } f(x, y) = \int Q(x, y) dy + h(x), \text{ where } h \text{ is independent of } y.$$

$$f(x, y) = \int P(x, y) dx + g(y) = x^3y + g(y) = \int Q(x, y) dy + h(x) = x^3y + h(x),$$

suggests that h, g should be such that, $h = g = c$.

Hence F is the gradient vector of any f of the form $f(x, y) = x^3y + c$.

(b) Since P, Q are continuously differentiable functions, (where $P(x, y) = \sin y - y \sin x + x$ and $Q(x, y) = \cos x + x \cos y + y$)

and $\frac{\partial P}{\partial y} = \cos y - \sin x = \frac{\partial Q}{\partial x}$, F is the gradient of some scalar function f .

Note that if f is such that F is the gradient vector of f , then

$$\frac{\partial f}{\partial x} = P(x, y) \quad \text{and} \quad \frac{\partial f}{\partial y} = Q(x, y) \quad \text{hence}$$

$$f(x, y) = \int P(x, y) dx + g(y), \text{ where } g \text{ is independent of } x.$$

Also $f(x, y) = \int Q(x, y) dy + h(x)$, where h is independent of y . Since

$$\int P(x, y) dx + g(y) = x \sin y + y \cos x + \frac{x^2}{2} + g(y) = \int Q(x, y) dy + h(x) = y \cos x + x \sin y + \frac{y^2}{2} + h(x),$$

if we choose $g(y) = \frac{y^2}{2}$ and $h(x) = \frac{x^2}{2}$,

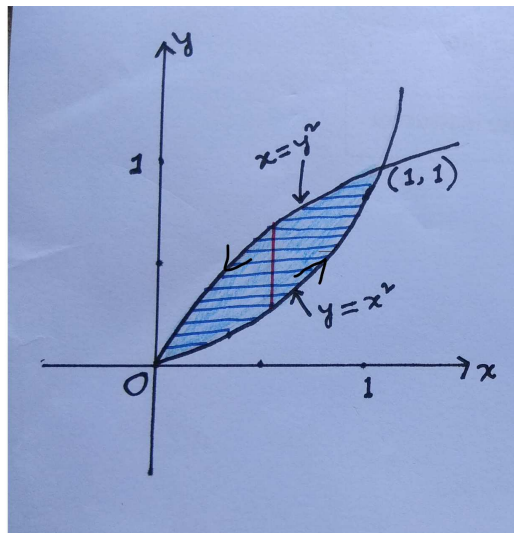
then $f(x, y) = x \sin y + y \cos x + \frac{x^2}{2} + \frac{y^2}{2}$ is such that $\nabla f = F$.

5. Use Green's theorem to evaluate the line integral along the given positively oriented curve:

- (a) $\int_C (y + e^{\sqrt{x}})dx + (2x + \cos y^2)dy$, C is the boundary of the region enclosed by the parabolas $y = x^2$ and $x = y^2$.

Solution: By Green's theorem:

$$\begin{aligned} \int_C (y + e^{\sqrt{x}})dx + (2x + \cos y^2)dy &= \int_0^1 \int_{x^2}^{\sqrt{x}} \left(\frac{\partial}{\partial x}(2x + \cos y^2) - \frac{\partial}{\partial y}(y + e^{\sqrt{x}}) \right) dydx \\ &= \int_0^1 \int_{x^2}^{\sqrt{x}} (2 - 1) dydx \\ &= \frac{1}{3}. \end{aligned}$$



- (b) $\int_C xydx + 2x^2dy$, C consists of the line segment from $(-2, 0)$ to $(2, 0)$ and top half of the circle $x^2 + y^2 = 4$.

Solution: By Green's theorem:

$$\begin{aligned} \int_C xydx + 2x^2dy &= \int_{-2}^2 \int_0^{\sqrt{4-x^2}} \left(\frac{\partial}{\partial x}(2x^2) - \frac{\partial}{\partial y}(xy) \right) dydx \\ &= \int_{-2}^2 \int_0^{\sqrt{4-x^2}} (4x - x) dydx \quad (*) \\ &= \int_{-2}^2 \sqrt{4-x^2} 3xdx = \frac{3}{2} \left[-\frac{2}{3}(4-x^2)^{\frac{3}{2}} \right]_{-2}^2 = 0. \end{aligned}$$

Aliter: By converting $(*)$ to polar coordinates we get:

$$\begin{aligned} &= \int_0^\pi \int_0^2 3r \cos \theta r dr d\theta \\ &= 0. \end{aligned}$$

6. Use Green's theorem to find out the work done by the force $\mathbf{F}(x, y) = x(x + y)\mathbf{i} + xy^2\mathbf{j}$ in moving a particle from the origin along x -axis to $(1, 0)$ and then along a line segment to $(0, 1)$, and then back to the origin along y -axis.

Solution: The work done is given by $\int_C \mathbf{F} \cdot d\mathbf{r}$, where C is the given curve.

$$\begin{aligned} \text{By Green's theorem } \int_C \mathbf{F} \cdot d\mathbf{r} &= \int_0^1 \int_0^{-x+1} \left(\frac{\partial}{\partial x}(xy^2) - \frac{\partial}{\partial y}(x(x + y)) \right) dydx \\ &= \int_0^1 \int_0^{-x+1} (y^2 - x) dydx \\ &= -\frac{1}{12}. \end{aligned}$$

7. Let D be a region bounded by a simple closed path C in the xy -plane. Use Green's theorem to prove that the coordinates of the centroid $(\bar{x}, \bar{y})$ of D are

$$\bar{x} = \frac{1}{2A} \oint x^2 dy, \quad \bar{y} = -\frac{1}{2A} \oint y^2 dx \quad \text{where } A \text{ is the area of } D.$$

Solution: By definition $\bar{x} = \frac{\iint_D x dA}{A}$, $\bar{y} = \frac{\iint_D y dA}{A}$, where $A = \text{Area}(D)$.

But $\iint_D x dA = \iint_D \left(\frac{\partial}{\partial x} \left(\frac{1}{2} x^2 \right) - \frac{\partial}{\partial y} (0) \right) dA \quad (P = 0, Q = \frac{1}{2} x^2).$

By Green's theorem $\iint_D x dA = \frac{1}{2} \oint_C x^2 dy$, hence $\bar{x} = \frac{1}{2A} \oint x^2 dy$.

Similarly $\iint_D y dA = \iint_D \left(\frac{\partial}{\partial x} (0) - \frac{\partial}{\partial y} \left(\frac{1}{2} y^2 \right) \right) dA \quad (Q = 0, P = \frac{1}{2} y^2).$

By Green's theorem $\iint_D y dA = \frac{1}{2} \oint_C y^2 dx$, hence $\bar{y} = \frac{1}{2A} \oint y^2 dx$.

8. If $\mathbf{r}(x, y) = x\mathbf{i} + y\mathbf{j}$ and $r = |\mathbf{r}|$, let

$$f(x, y) = \frac{\partial(\log r)}{\partial y} \mathbf{i} - \frac{\partial(\log r)}{\partial x} \mathbf{j}$$

for $r > 0$.

Let C be a smooth simple closed curve in the annulus $1 < x^2 + y^2 < 25$, then find all possible values of the line integral of f along C .

Solution: Let C_1 denote the positively oriented unit circle centered at the origin.

Note that $f(x, y) = \frac{y}{(x^2 + y^2)} \mathbf{i} - \frac{x}{(x^2 + y^2)} \mathbf{j}$, for $(x, y) \neq (0, 0)$.

$\frac{\partial P}{\partial y}, \frac{\partial Q}{\partial x} \quad \left(= \frac{x^2 - y^2}{(x^2 + y^2)^2} \right) \quad \text{are continuous for all } (x, y) \in \mathbf{R}^2 \setminus \{(0, 0)\},$

where $P(x, y) = \frac{y}{(x^2 + y^2)}$ and $Q(x, y) = -\frac{x}{(x^2 + y^2)}$.

By Green's theorem $\iint_D \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) dA = 0 = \oint_C (P(x, y) dx + Q(x, y) dy),$

if C is such that the origin is not inside C .

By Green's theorem again

$\iint_D \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) dA = 0 = \oint_C (P(x, y) dx + Q(x, y) dy) - \oint_{C_1} (P(x, y) dx + Q(x, y) dy),$

if C is such that the origin is inside C , and C is positively oriented.

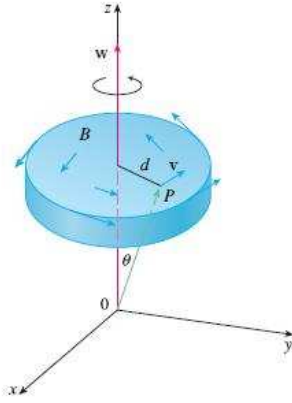
Hence $\oint_C (P(x, y) dx + Q(x, y) dy) = -2\pi$. (refer to (***) of problem 3)

$\iint_D \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) dA = 0 = -\oint_C (P(x, y) dx + Q(x, y) dy) - \oint_{C_1} (P(x, y) dx + Q(x, y) dy),$

if C is such that the origin is inside C , and C is negatively oriented.

Hence $\oint_C (P(x, y) dx + Q(x, y) dy) = 2\pi$.

Hence the only possible values of the line integral are $0, \pm 2\pi$.



9. The exercise demonstrates a connection between curl vector and rotations. Let $\mathbf{B}$ be a rigid body rotating about z -axis. The rotation can be described by the vector $\mathbf{w} = \omega \mathbf{k}$, where ω is the angular speed of $\mathbf{B}$, that is, the tangential speed at any point P in B divided by the distance d from the axis of rotation. Let $\mathbf{r} = \langle x, y, z \rangle$ be the position vector of P .

- (a) By considering the angle θ in the figure, show that the velocity field of B is given by $\mathbf{v} = \mathbf{w} \times \mathbf{r}$.
 (b) Show that $\mathbf{v} = -\omega y \mathbf{i} + \omega x \mathbf{j}$.
 (c) Show that $\nabla \times \mathbf{v} = 2\mathbf{w}$.

Solution:

(a) Given that $\omega = \frac{|\mathbf{v}|}{d}$, and $\sin \theta = \frac{d}{|\mathbf{r}|}$.

Therefore $|\mathbf{v}| = d\omega = \omega|\mathbf{r}|\sin \theta = |\mathbf{w} \times \mathbf{r}|$. (1)

Since $\mathbf{v}$ is orthogonal to both $\mathbf{w}$ and $\mathbf{r}$ and its direction is same as that of $\mathbf{w} \times \mathbf{r}$ therefore from (1) it follows:

$\mathbf{v} = \mathbf{w} \times \mathbf{r}$.

(b) Since $\mathbf{w} = \omega \mathbf{k}$ and $\mathbf{r}(x, y, z) = x\mathbf{i} + y\mathbf{j} + z\mathbf{k}$,

$$\mathbf{v} = \mathbf{w} \times \mathbf{r} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 0 & 0 & \omega \\ x & y & z \end{vmatrix} = -\omega y \mathbf{i} + \omega x \mathbf{j}.$$

$$(c) \nabla \times \mathbf{v} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ -\omega y & \omega x & 0 \end{vmatrix} = 2\mathbf{w}.$$

10. Use Stokes' Theorem to evaluate

- (a) $\iint_S (\nabla \times \mathbf{F}) \cdot d\mathbf{S}$ where $\mathbf{F} = xyz\mathbf{i} + xy\mathbf{j} + x^2yz\mathbf{k}$ and S consists of the top and the four sides (but not the bottom) of the cube with vertices $(\pm 1, \pm 1, \pm 1)$, oriented outward.

$$\text{Solution: } \iint_S (\nabla \times \mathbf{F}) \cdot d\mathbf{S} = \iint_S (\nabla \times \mathbf{F}) \cdot \mathbf{n} dS.$$

$$\text{By Stokes' theorem, } \iint_S (\nabla \times \mathbf{F}) \cdot d\mathbf{S} = \int_{C_1} Pdx + Qdy + Rdz + \int_{C_2} Pdx + Qdy + Rdz$$

$$+ \int_{C_3} Pdx + Qdy + Rdz + \int_{C_4} Pdx + Qdy + Rdz$$

where $C_i, i = 1, 2, \dots, 4$ are the four sides of the bottom surface of the cube, and $P(x, y, z) = xyz$, $Q(x, y, z) = xy$, $R(x, y, z) = x^2yz$.

Since for $C_1, y = -1, z = -1$: For $C_2, x = 1, z = -1$: For $C_3, y = 1, z = -1$: For $C_4, x = -1, z = -1$,

$$\int_{C_1} Pdx + Qdy + Rdz + \int_{C_2} Pdx + Qdy + Rdz + \int_{C_3} Pdx + Qdy + Rdz$$

$$+ \int_{C_4} Pdx + Qdy + Rdz$$

$$= \int_{C_1} Pdx + \int_{C_2} Qdy + \int_{C_3} Pdx + \int_{C_4} Qdy$$

$$= \int_{-1}^1 (-1)(-1)xdx + \int_{-1}^1 (-1)(1)ydy + \int_1^{-1} (-1)(1)xdx + \int_1^{-1} (-1)(-1)ydy = 0.$$

- (b) $\int_C \mathbf{F} \cdot d\mathbf{r}$ where $\mathbf{F} = 2z\mathbf{i} + 4x\mathbf{j} + 5y\mathbf{k}$ and C is the curve of intersection of the plane $z = x + 4$ and the cylinder $x^2 + y^2 = 4$.

Solution: $\int_C \mathbf{F} \cdot d\mathbf{r} = \iint_S (\nabla \times \mathbf{F}) \cdot \mathbf{n} dS$.

$$\nabla \times \mathbf{F} = \left(\frac{\partial R}{\partial y} - \frac{\partial Q}{\partial z}\right)\mathbf{i} + \left(\frac{\partial P}{\partial z} - \frac{\partial R}{\partial x}\right)\mathbf{j} + \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}\right)\mathbf{k} = 5\mathbf{i} + 2\mathbf{j} + 4\mathbf{k},$$

where $P(x, y, z) = 2z$, $Q(x, y, z) = 4x$, $R(x, y, z) = 5y$.

The surface $\mathbf{r}(x, y)$ is of the form: $\mathbf{r}(x, y) = x\mathbf{i} + y\mathbf{j} + f(x, y)\mathbf{k}$, where $f(x, y) = x + 4$, hence

$$\frac{\partial \mathbf{r}}{\partial x} \times \frac{\partial \mathbf{r}}{\partial y} = -\frac{\partial f}{\partial x}\mathbf{i} - \frac{\partial f}{\partial y}\mathbf{j} + \mathbf{k} = -\mathbf{i} + \mathbf{k}$$

$$\iint_S (\nabla \times \mathbf{F}) \cdot \mathbf{n} dS = \iint_D (5\mathbf{i} + 2\mathbf{j} + 4\mathbf{k}) \cdot (-\mathbf{i} + \mathbf{k}) dA,$$

where D is the disc of radius 2 centered at the origin.

$$\text{Hence } \iint_S (\nabla \times \mathbf{F}) \cdot \mathbf{n} dS = \iint_D (-5 + 4) dA = -\int_0^2 \int_0^{2\pi} r d\theta dr = -4\pi.$$

11. Calculate the work done by the force field

$$\mathbf{F}(x, y, z) = (x^x + z^2)\mathbf{i} + (y^y + x^2)\mathbf{j} + (z^z + y^2)\mathbf{k}$$

when a particle moves under its influence around the edge of the part of the sphere $x^2 + y^2 + z^2 = 4$ that lies in the first octant, in a counterclockwise direction as viewed from above.

Solution: The work done is given by

$$\int_C \mathbf{F} \cdot d\mathbf{r} = \iint_S (\nabla \times \mathbf{F}) \cdot \mathbf{n} dS.$$

$$\left(\frac{\partial R}{\partial y} - \frac{\partial Q}{\partial z}\right)\mathbf{i} + \left(\frac{\partial P}{\partial z} - \frac{\partial R}{\partial x}\right)\mathbf{j} + \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}\right)\mathbf{k} = 2(y\mathbf{i} + z\mathbf{j} + x\mathbf{k}).$$

The surface $\mathbf{r}(x, y)$ is of the form $\mathbf{r}(x, y) = x\mathbf{i} + y\mathbf{j} + f(x, y)\mathbf{k}$ where $f(x, y) = \sqrt{4 - x^2 - y^2}$, hence

$$\frac{\partial \mathbf{r}}{\partial x} \times \frac{\partial \mathbf{r}}{\partial y} = -\frac{\partial f}{\partial x}\mathbf{i} - \frac{\partial f}{\partial y}\mathbf{j} + \mathbf{k} = \frac{x}{\sqrt{4 - x^2 - y^2}}\mathbf{i} + \frac{y}{\sqrt{4 - x^2 - y^2}}\mathbf{j} + \mathbf{k}$$

$$\iint_S (\nabla \times \mathbf{F}) \cdot \mathbf{n} dS = \iint_D 2(y\mathbf{i} + z\mathbf{j} + x\mathbf{k}) \cdot \left(\frac{x}{\sqrt{4 - x^2 - y^2}}\mathbf{i} + \frac{y}{\sqrt{4 - x^2 - y^2}}\mathbf{j} + \mathbf{k}\right) dA,$$

where D is the part of the disc of radius 2 centered at the origin in the first quadrant.

$$\text{Hence } \iint_S (\nabla \times \mathbf{F}) \cdot \mathbf{n} dS = 2 \iint_D \left(x + y + \frac{xy}{\sqrt{4 - x^2 - y^2}}\right) dA$$

$$= 2 \int_0^2 \int_0^{\frac{\pi}{2}} \left((r \cos \theta + r \sin \theta) + \left(\frac{r^2 \cos \theta \sin \theta}{\sqrt{4 - r^2}}\right)\right) r d\theta dr = 2(I_1 + I_2) = 2\left(\frac{16}{3} + \frac{8}{3}\right) = 16.$$

12. Let S be the surface of the solid cylinder T bounded by the planes $z = 0$ and $z = 3$ and the cylinder $x^2 + y^2 = 4$. Calculate the outward flux $\iint_S \mathbf{F} \cdot \mathbf{n} dS$ given $\mathbf{F}(x, y, z) = (x^2 + y^2 + z^2)(x\mathbf{i} + y\mathbf{j} + z\mathbf{k})$.

Solution: By divergence theorem,

$$\iint_S \mathbf{F} \cdot \mathbf{n} dS = \iiint_T \text{Div}(\mathbf{F}) dv.$$

$$\text{Div}(\mathbf{F}) = \nabla \cdot \mathbf{F} = \frac{\partial P}{\partial x} + \frac{\partial Q}{\partial y} + \frac{\partial R}{\partial z} = 5(x^2 + y^2 + z^2),$$

where $P = (x^2 + y^2 + z^2)x$, $Q = (x^2 + y^2 + z^2)y$ and $R = (x^2 + y^2 + z^2)z$.

By using cylindrical coordinates we get:

$$\iint_S \mathbf{F} \cdot \mathbf{n} dS = \iiint_T \text{Div}(\mathbf{F}) dv = \int_0^3 \int_0^{2\pi} \int_0^2 5(r^2 + z^2) r dr d\theta dz = 300\pi.$$

13. Use Divergence Theorem to evaluate $\iint_S \mathbf{F} \cdot \mathbf{n} dS$ where

$$\mathbf{F}(x, y, z) = z^2 x \mathbf{i} + \left(\frac{1}{3} y^3 + \tan z \right) \mathbf{j} + (x^2 z + y^2) \mathbf{k}$$

and S is the top half of the sphere $x^2 + y^2 + z^2 = 1$.

Solution: Let S_1 be the unit disc centered at the origin with normal facing downward, and let $S_2 = S \cup S_1$, then S_2 has outer facing normal if S has upward facing normal.

$$\iint_{S_2} \mathbf{F} \cdot \mathbf{n} dS = \iint_S \mathbf{F} \cdot \mathbf{n} dS + \iint_{S_1} \mathbf{F} \cdot \mathbf{n} dS.$$

$$\text{By divergence theorem } \iint_{S_2} \mathbf{F} \cdot \mathbf{n} dS = \iiint_T \text{Div}(F) dv.$$

$$\text{Since } \text{Div}(F) = \nabla \cdot F = \frac{\partial P}{\partial x} + \frac{\partial Q}{\partial y} + \frac{\partial R}{\partial z} = z^2 + y^2 + x^2,$$

$$\text{where } P = z^2 x, \quad Q = \frac{1}{3} y^3 + \tan z, \quad R = x^2 z + y^2.$$

By using spherical coordinates we get:

$$\iint_{S_2} \mathbf{F} \cdot \mathbf{n} dS = \int_0^1 \int_0^{2\pi} \int_0^{\frac{\pi}{2}} \rho^2 (\rho^2 \sin \phi) d\phi d\theta d\rho = \frac{2\pi}{5}.$$

$$\text{Also } \iint_{S_1} \mathbf{F} \cdot \mathbf{n} dS = \iint_{S_1} \mathbf{F} \cdot (-\mathbf{k}) dS = \iint_{S_1} (x^2 \times 0 + y^2)(-1) dS = - \int_0^1 \int_0^{2\pi} (r^2 \sin^2 \theta) r dr d\theta = -\frac{\pi}{4}$$

$$\iint_S \mathbf{F} \cdot \mathbf{n} dS = \iint_{S_2} \mathbf{F} \cdot \mathbf{n} dS - \iint_{S_1} \mathbf{F} \cdot \mathbf{n} dS = \frac{13\pi}{20}.$$